Session 2 — Student Reading

Spatial Analysis of LMPs and Generation Siting

LaPSEE Minicourse

Introduction to Power Networks as Intelligent Graphs

Abstract

This reading accompanies Session 2 of the LaPSEE minicourse and prepares you for the lab notebook `s02_spatial-prices-siting.ipynb`. We answer a single question — *why do two adjacent buses see different energy prices?* — and follow its consequences all the way to a generation-siting decision. The locational marginal price (LMP) is derived as the dual variable of the DC-OPF nodal balance constraint (Schweppe et al. 1988; Hogan 1992; Litvinov 2010); its spatial structure is shown to be *mechanically* produced by the PTDF matrix restricted to binding lines. We then bring the spatial-econometrics machinery of Session 1 (Moran's I, LISA, SAR/SEM/SDM, the LeSage–Pace effects decomposition) to bear on a 200-hour LMP panel, fit a siting model on the IEEE 118 system, and sanity-check everything on the realistic 73-bus RTS-GMLC benchmark. Worked examples use the IEEE 30, IEEE 118, and RTS-GMLC systems; every numerical value in the text is reproduced by an executable cell in the notebook (`RANDOM_SEED`=2026). The four primary references for this session are Schweppe et al. (1988), J. LeSage and Pace (2009), Anselin, Bera, et al. (1996), and Faria et al. (2016).

Learning objectives

By the end of this reading and notebook you will be able to:

1. State the DC-OPF as a convex program and *derive* the LMP as the dual of the nodal balance constraint via the KKT conditions.
2. Decompose $\text{LMP}_i = \lambda_{\text{energy}} + \text{LMP}_i^{\text{cong}} (+\text{LMP}_i^{\text{loss}})$ and explain why the congestion term $-\sum_\ell \mu_\ell H_{\ell,i}$ is the source of spatial structure.
3. Choose, with justification, among four spatial weight matrices $\mathbf{W}$ for a power network, and argue why the PTDF-derived $\mathbf{W}$ is principled for LMP analysis.
4. Compute Moran's I and LISA on a nodal-price panel and relate the spatial autocorrelation to the congestion regime.
5. Estimate SAR, SEM, and SDM by maximum likelihood, run the Anselin, Bera, et al. (1996) LM-diagnostic battery to select a specification, and report direct/indirect/total effects rather than raw $\hat{\boldsymbol{\beta}}$.
6. Frame distributed-generation siting as a spatial regression and rank candidate buses by the *average total effect*, internalising network spillovers.
7. Explain, using the RTS-GMLC contrast, why the strength of spatial dependence in LMPs is an empirical property of the benchmark, not a given.

Contents

1 Why prices vary across the grid

Motivation. In an uncongested, lossless power system every bus clears at the same uniform energy price: electricity is a commodity and the network is invisible. The moment a single line constraint binds, that symmetry breaks. The marginal cost of serving one more megawatt of demand now depends on *where* that demand sits relative to the binding constraint, because the cheapest feasible re-dispatch differs bus by bus. Schweppe et al. (1988) formalised this insight as *spot pricing*: the price at a bus is the Lagrange multiplier (shadow price) of that bus's power-balance constraint in the optimal dispatch. Hogan (1992) gave the financial-transmission-rights interpretation that underpins modern ISO markets, and Litvinov (2010) provides the clean contemporary exposition we follow.

The pedagogical claim of this session is that the spatial structure of nodal prices is not a geographic accident to be modelled with distance-decay weights — it is *mechanically determined* by the grid's physics (the PTDF matrix) and by which constraints bind. Once we see that, the spatial-econometrics toolbox of Session 1 becomes the natural language for describing, testing, and exploiting that structure.

2 The DC-OPF and the LMP decomposition

2.1 DC power flow and PTDF (Session 1 recap)

Recall from Session 1 the lossless DC power-flow model. With $\mathbf{B}$ the susceptance Laplacian, nodal injections P^{inj}, and bus angles $\boldsymbol{\theta}$,

$$P^{\text{inj}} = -\mathbf{B}\,\boldsymbol{\theta}, \qquad \boldsymbol{\theta}_{-s} = \mathbf{B}^{-1}_{-s,-s}\,P^{\text{inj}}_{-s}, \quad \theta_s = 0,$$

where s is the slack bus. Writing $\mathbf{B}_r \equiv \mathbf{B}_{-s,-s}$ for the reduced susceptance Laplacian, $\mathbf{B}_d = \text{diag}(b_\ell)$ for the diagonal of branch susceptances, and $\mathbf{A}_r$ for the reduced branch–bus incidence matrix, the vector of branch flows is $\boldsymbol{f} = \mathbf{B}_d\,\mathbf{A}_r\,\boldsymbol{\theta}_{-s}$. Substituting the angle solution gives the *Power Transfer Distribution Factor* matrix.

Proposition 2.1 (PTDF closed form). *Under the DC assumptions above, branch flows are linear in nodal injections,*

$$\boldsymbol{f} = \mathbf{H}\,\boldsymbol{p}, \qquad \boxed{\mathbf{H} = \mathbf{B}_d\,\mathbf{A}_r\,\mathbf{B}_r^{-1}}$$

where $\boldsymbol{p} \in \mathbb{R}^{n-1}$ *is the vector of non-slack net injections. Row* ℓ *of* $\mathbf{H}$ *lists the sensitivities of the flow on line* ℓ *to a unit injection at each non-slack bus, balanced by the slack.* $\mathbf{H}$ *depends only on the network topology and reactances — not on the dispatch.*

Proof sketch. From $\boldsymbol{p} = \mathbf{B}_r\,\boldsymbol{\theta}_{-s}$ we get $\boldsymbol{\theta}_{-s} = \mathbf{B}_r^{-1}\boldsymbol{p}$; substitute into $\boldsymbol{f} = \mathbf{B}_d\,\mathbf{A}_r\,\boldsymbol{\theta}_{-s}$. The slack column is zero by construction because an injection there is fully absorbed by the reference. See Wang et al. (2011) for the same object used to build electrical betweenness. □ □

2.2 The DC-OPF as a convex program

Let generators have quadratic costs $C_g(P_g) = a_g + b_g P_g + \frac{1}{2}c_g P_g^2$ and demand $\boldsymbol{P^d}$ be exogenous. Writing $\Delta\boldsymbol{P} = \boldsymbol{P}^{\boldsymbol{g}}_{\text{net}} - \boldsymbol{P}^{\boldsymbol{d}}_{\text{net}}$ for the vector of net nodal injections, the lossless DC optimal power flow is

$$\begin{aligned}
\min_{\boldsymbol{P^g}} \ & \sum_g C_g(P_g) \\
\text{s.t. } \mathbf{1}^\top \boldsymbol{P^g} - \mathbf{1}^\top \boldsymbol{P^d} &= 0 \quad (\lambda) \text{ [system power balance]} \\
\pm\, \mathbf{H}\,\Delta\boldsymbol{P} &\le \boldsymbol{F}^{\max} \quad (\boldsymbol{\mu}^{\pm} \ge \mathbf{0}) \text{ [line limits]} \\
P_g^{\min} \le P_g &\le P_g^{\max} \quad (\underline{\boldsymbol{\nu}}, \overline{\boldsymbol{\nu}} \ge \mathbf{0}).
\end{aligned}$$

Quadratic costs plus linear constraints make the program convex, so the Karush–Kuhn–Tucker (KKT) conditions are necessary and sufficient.

Proposition 2.2 (LMP decomposition, lossless DC-OPF). *At an optimum of the DC-OPF, the locational marginal price at bus* i *— defined as the sensitivity of the optimal cost to demand at* i *— is*

$$\boxed{LMP_i \;=\; \underbrace{\lambda}_{energy} \;-\; \underbrace{\sum_\ell \mu_\ell\, H_{\ell,i}}_{congestion}} \qquad \mu_\ell \equiv \mu_\ell^+ - \mu_\ell^-,$$

where λ *is the dual of the system power balance and* $\mu_\ell \ge 0$ *are the duals of the line-limit constraints. The slack column of* $\mathbf{H}$ *is zero, so* $LMP_s = \lambda$ *identifies the energy component.*

Proof sketch. Form the Lagrangian $\mathcal{L} = \sum_g C_g(P_g) - \lambda(\mathbf{1}^\top\boldsymbol{P^g} - \mathbf{1}^\top\boldsymbol{P^d}) + \boldsymbol{\mu}^{+\top}(\mathbf{H}\Delta\boldsymbol{P} - \boldsymbol{F}^{\max}) + \dots$ Stationarity in an interior generator g at bus i gives $C_g'(P_g) = \lambda - \sum_\ell \mu_\ell\, H_{\ell,i}$. The marginal cost of serving demand at i equals the marginal generation cost that the optimal dispatch is willing to incur there, i.e. $\text{LMP}_i = -\,\partial\mathcal{L}^\star/\partial P_i^d = \lambda - \sum_\ell \mu_\ell\, H_{\ell,i}$. The full KKT derivation is given in the professor reading; see Litvinov (2010). □ □

Remark. Adding a quadratic loss term $P_{\text{loss}}(\boldsymbol{\theta}) \approx \frac{1}{2}\boldsymbol{\theta}^\top\mathbf{G}\boldsymbol{\theta}$ to the balance constraint introduces a third component, $\text{LMP}_i = \lambda(1+\delta_i) - \sum_\ell \mu_\ell H_{\ell,i}$ with $\delta_i = \partial P_{\text{loss}}/\partial P_i$ the marginal loss factor (Litvinov 2010). The loss term is typically ±1–5%; the lab uses the lossless approximation, so we work with the two-term decomposition.

Example 2.1 (IEEE 30 single OPF snapshot, notebook §1). We rate the IEEE 30 backbone realistically (50 MVA on the 132 kV lines, 16 MVA on the 33 kV ring), solve the DC-OPF with the slack at bus 0, and decompose the nodal prices. The objective is 9595.92 € and exactly two constraints bind (line 0, bus $0 \to 1$, and line 20, bus $9 \to 20$). Because the slack PTDF column is zero, $\lambda_{\text{energy}} = \text{LMP}_0 = 26.83$ €/MWh, and the LMP spread is 17.765 €/MWh:

bus	LMP_i (€/MWh)	λ_{energy}	congestion
0 (slack)	26.830	26.830	0.000
1	44.595	26.830	**+17.765**
2	37.075	26.830	+10.244
3	39.425	26.830	+12.594
4	42.627	26.830	+15.796

The buses with the largest congestion premium are exactly those whose PTDF column "lights up" a binding line: bus 1 has $\max_\ell |H_{\ell,1}| = 0.833$ on line 0, and bus 20 has 0.575 on line 20. The notebook self-check asserts that the congestion component of the slack bus is zero.

2.3 Why LMPs carry spatial structure

Stacking the per-bus prices into $\boldsymbol{p}^{\text{LMP}} \in \mathbb{R}^n$ and using Proposition 2.2 with the loss term,

$$\boldsymbol{p}^{\text{LMP}} = \lambda\,\mathbf{1} + \lambda\,\boldsymbol{\delta} - \mathbf{H}^\top \boldsymbol{\mu}.$$

Three observations follow, and they are the entire reason spatial econometrics applies here:
1. $\lambda\,\mathbf{1}$ is spatially constant — it contributes nothing to autocorrelation.
2. $\lambda\,\boldsymbol{\delta}$ is smooth in network position, inducing *positive* spatial dependence.
3. $-\mathbf{H}^\top\boldsymbol{\mu}$ is discrete and structured by which lines bind; it can flip sign across a binding line, producing *negative local* autocorrelation that LISA can detect (§4).

In a stressed grid only a small set $\mathcal{B}$ of lines bind, so the covariance kernel of nodal prices is approximately $\sum_{\ell\in\mathcal{B}} \mathbf{H}_{\ell,:}^\top \mathbf{H}_{\ell,:}$ — a sum of rank-one operators built from PTDF rows. Any $\mathbf{W}$ that approximates this kernel captures the coupling *mechanism*; a geographic-distance $\mathbf{W}$ is at best a proxy. This motivates the menu of weight matrices in the next section.

3 Four spatial weight matrices for a power network

Motivation. In spatial econometrics $\mathbf{W}$ encodes the researcher's prior about which units interact (Anselin and Rey 2014). Session 1 introduced four credible choices for a power grid; here we recall them and ask which is principled for LMP analysis.
1. **Binary adjacency $\mathbf{W} = \mathbf{A}$** (first-order contiguity): does a line exist between i and j?
2. **Electrical distance $\mathbf{W}_{ij} = 1/Z_{\text{eff}}(i,j)$**, where Z_{eff} is the effective-impedance (resistance) distance (Klein and Randić 1993; Dörfler et al. 2018).
3. **PTDF-thresholded $\mathbf{W}_{ij}$** $= \max_\ell |H_{\ell,i} - H_{\ell,j}|$, retaining the top-$q$ quantile of entries.
4. **Geographic k-nearest-neighbours** in bus coordinates (x, y).

Throughout we set $W_{ii} = 0$, row-standardise so that $\tilde{\mathbf{W}}\,\mathbf{1} = \mathbf{1}$, and treat $\mathbf{W}$ as exogenous and known.

Example 3.1 (IEEE 30 sparsity comparison, notebook §1b / §2a.2)**.** On IEEE 30 the four constructions differ enormously in density yet share the same post-standardisation spectral radius:

$\mathbf{W}$ choice	non-zeros	density	max entry	$\mu_{\max}(\tilde{\mathbf{W}})$
$\mathbf{W} = \mathbf{A}$	82	0.091	1.000	1.000
$\mathbf{W}_{\text{elec}} = 1/Z_{\text{eff}}$	870	0.967	47.47	1.000
$\mathbf{W}_{\text{PTDF}}$ (top 25%)	218	0.242	1.000	1.000
$\mathbf{W}_{k\text{NN}}$ ($k = 4$)	144	0.160	1.000	1.000

Because every row-standardised $\mathbf{W}$ has $\mu_{\max} = 1$, the admissible SAR range $|\rho| < 1$ is identical across choices; what changes is the *interpretation* of ρ. The electrical-distance matrix is nearly dense (96.7% non-zero), which, as Exercise 10.5 shows, can push the SAR estimate to the stationarity boundary.

Lemma 3.1 (Spatial multiplier and Neumann series)**.** *If* $|\rho| < 1/|\mu_{\max}(\mathbf{W})|$ *then* $(\mathbf{I} - \rho\mathbf{W})$ *is invertible and the spatial multiplier admits the expansion*

$$\mathbf{S}_\rho \equiv (\mathbf{I} - \rho\mathbf{W})^{-1} = \sum_{k\geq 0} \rho^k\, \mathbf{W}^k.$$

For a row-standardised $\mathbf{W}$ *this holds whenever* $|\rho| < 1$*, and* $\mathbf{S}_\rho\,\mathbf{1} = \frac{1}{1-\rho}\mathbf{1}$.

Proof sketch. $\|\rho\mathbf{W}\| < 1$ under the hypothesis, so the partial sums of the geometric series telescope to $(\mathbf{I} - \rho\mathbf{W})^{-1}$. The entry $(\mathbf{W}^k)_{ij}$ counts weighted walks of length k from j to i, so $\mathbf{S}_\rho$ aggregates spillovers discounted geometrically by ρ. $\square$ $\square$

Key idea

For LMP analysis the PTDF-derived $\mathbf{W}$ is the principled main specification because it mirrors the coupling mechanism of §2.3; the binary adjacency $\mathbf{W} = \mathbf{A}$ is the natural robustness check. A defensible study *declares which* $\mathbf{W}$ *it used and why*, and reports at least one alternative (LeSage's misspecification argument, J. P. LeSage 2008).

4 Moran's I and LISA on nodal prices

Global Moran's I. Centre the price vector, $\boldsymbol{z} = \boldsymbol{p}^{\text{LMP}} - \bar{p}^{\text{LMP}}\mathbf{1}$. The global Moran statistic measures whether neighbouring buses (under $\mathbf{W}$) have similar prices:

$$I = \frac{n}{S_0}\,\frac{\boldsymbol{z}^\top \mathbf{W}\boldsymbol{z}}{\boldsymbol{z}^\top\boldsymbol{z}}, \qquad S_0 = \sum_{i,j} W_{ij}.$$

Under the null of no spatial association $\mathbb{E}[I] = -1/(n-1) \to 0$; because the LMP distribution is heavy-tailed and n is small (IEEE 30: $n = 30$), inference uses 999 random permutations of $\boldsymbol{z}$ rather than normal theory (Moran 1950; Anselin 1995).

Example 4.1 (Moran's I on the IEEE 30 LMP panel, notebook §2a.2–2a.3)**.** A stylized daily load profile (peak 1.02×, trough 0.65× base, mild noise) drives a 200-hour DC-OPF panel; 199 hours are feasible, mean LMP 39.43 €/MWh. Computing I hour-by-hour under two weight matrices:

$\mathbf{W}$ choice	mean I across hours	std dev
$\mathbf{W} = \mathbf{A}$	**+0.066**	0.048
$\mathbf{W} = \mathbf{W}_{\text{PTDF}}$	−0.005	0.005

The headline result is dynamic, not static: the correlation between $|I_t|$ and the number of binding lines in hour t is $+$**0.867**. When no line binds the price map collapses to $\lambda\mathbf{1}$ and $I \to 0$; spatial autocorrelation in prices *tracks the congestion regime* (Figure 1).

Local Moran (LISA). The global statistic can hide local clusters that cancel. The Local Indicator of Spatial Association (Anselin 1995) decomposes I into per-bus contributions,

$$I_i = \frac{z_i}{m_2}\sum_j W_{ij} z_j, \qquad m_2 = \tfrac{1}{n}\sum_k z_k^2.$$

A significant I_i together with the quadrant of $(z_i, (\tilde{\mathbf{W}}\boldsymbol{z})_i)$ classifies bus i as HH, LL, HL, or LH. In power-systems terms, *High–High* clusters at peak hours are load pockets downstream of congestion, and *Low–Low* clusters are export-rich regions upstream. The notebook draws the Moran scatterplot and the LISA cluster map on the bus layout for the peak-congestion hour (cell §2a.7).

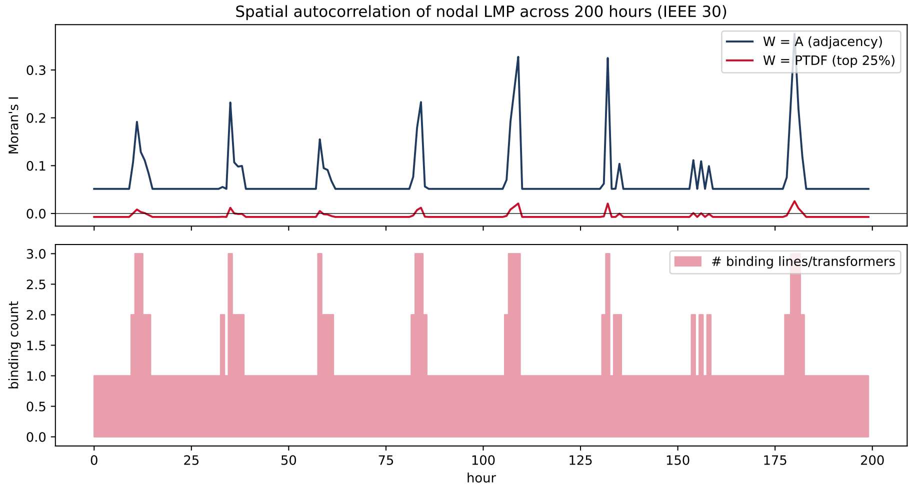

Figure 1: Top: global Moran's I of nodal LMPs across 200 hours on IEEE 30 under $\mathbf{W} = \mathbf{A}$ (navy) and $\mathbf{W} = \mathbf{W}_{\mathrm{PTDF}}$ (red). Bottom: the number of binding lines/transformers per hour. Spatial autocorrelation spikes exactly when constraints bind. Notebook §2a.3.

Example 4.2 (LISA on the peak-congestion hour, notebook §2a.7)**.** Selecting the hour with the largest $|I_t|$, standardising the LMP vector, and running `Moran_Local` with 999 permutations (`seed`=2026) produces Figure 2. The Moran scatterplot's slope equals the global I for that hour; the cluster map colours each bus by its HH/LL/HL/LH class at the 5% level. The High–High cluster is the load pocket downstream of the binding line; the Low–Low cluster is the export-rich region upstream; the HL/LH spatial outliers sit on the two sides of a binding line — the local signature of the discrete $-\mathbf{H}^\top\boldsymbol{\mu}$ term from §2.3.

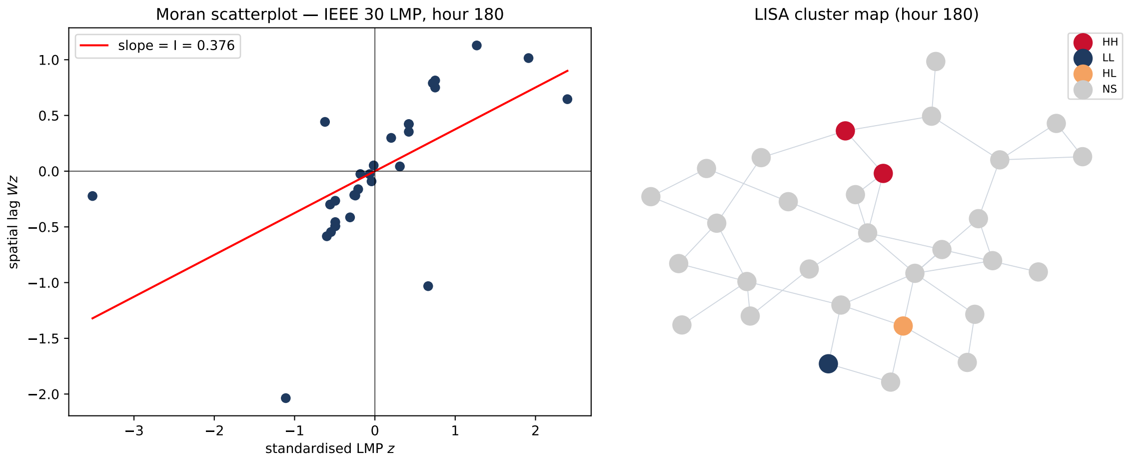

Figure 2: Left: Moran scatterplot of standardised LMPs at the peak-congestion hour (slope = I). Right: LISA cluster map on the IEEE 30 bus layout. Notebook §2a.7.

5 Spatial regression: SAR, SEM, SDM

Detecting spatial dependence is necessary but not sufficient; we want a regression that accounts for it while estimating the marginal effects of covariates. With $\mathbf{W}$ row-standardised, $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, and $\mathbf{X}$ of full column rank including a constant:

Name	Equation	Spatial term
SAR	$\mathbf{y} = \rho\,\mathbf{W}\mathbf{y} + \mathbf{X}\beta + \varepsilon$	lag of $\mathbf{y}$
SEM	$\mathbf{y} = \mathbf{X}\beta + \mathbf{u},\ \mathbf{u} = \lambda\mathbf{W}\mathbf{u} + \varepsilon$	lag of $\mathbf{u}$
SDM	$\mathbf{y} = \rho\mathbf{W}\mathbf{y} + \mathbf{X}\beta + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \varepsilon$	lag of $\mathbf{y}$ *and* $\mathbf{X}$

The SAR reduced form is $\mathbf{y} = \mathbf{S}_\rho(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$. Only SAR/SDM admit a spillover interpretation of $\boldsymbol{\beta}$; SEM treats spatial dependence as nuisance error correlation.

Proposition 5.1 (Inconsistency of OLS for SAR)**.** *Under SAR with $|\rho| < 1$ and row-standardised $\mathbf{W}$, the regressor $\mathbf{W}\mathbf{y}$ is correlated with $\boldsymbol{\varepsilon}$, and the OLS estimator $\hat{\rho}_{\text{OLS}}$ does not converge to ρ as $n \to \infty$.*

Proof sketch. From the reduced form, $\mathbf{W}\mathbf{y} = \mathbf{W}\mathbf{S}_\rho(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$, so

$$\mathbb{E}\big[(\mathbf{W}\mathbf{y})^\top \boldsymbol{\varepsilon}\big] = \sigma^2 \operatorname{tr}(\mathbf{W}\mathbf{S}_\rho) \neq 0 \quad \text{whenever } \rho \neq 0.$$

The trace is non-zero on a non-degenerate $\mathbf{W}$, so the endogeneity of $\mathbf{W}\mathbf{y}$ does not vanish in the limit. Two consistent estimators restore convergence: maximum likelihood (which adds the Jacobian $\log|\mathbf{I} - \rho\mathbf{W}|$ to the log-likelihood) and 2SLS/GMM (which instruments $\mathbf{W}\mathbf{y}$ with $[\mathbf{X}, \mathbf{W}\mathbf{X}, \mathbf{W}^2\mathbf{X}, \dots]$, valid because they depend only on exogenous $\mathbf{X}$ and $\mathbf{W}$; see Kelejian and Prucha 1998; Lee 2004). □ □

Direct, indirect, and total effects. Because $\partial\mathbb{E}[\mathbf{y}]/\partial X_{\cdot,r} = \mathbf{S}_\rho\,\beta_r$ is an *$n \times n$ matrix*, the raw coefficient β_r is *not* the marginal effect. J. LeSage and Pace (2009) summarise it with three scalars:

$$\text{ADE}_r = \tfrac{1}{n}\operatorname{tr}(\mathbf{S}_\rho)\beta_r, \quad \text{ATE}_r = \tfrac{1}{n}\mathbf{1}^\top\mathbf{S}_\rho\mathbf{1}\,\beta_r = \frac{\beta_r}{1-\rho}, \quad \text{AIE}_r = \text{ATE}_r - \text{ADE}_r.$$

In SDM the indirect effect also picks up θ_r, so it is non-zero even when $\rho = 0$. SDM is the "safe default" because it nests SAR ($\boldsymbol{\theta} = \mathbf{0}$) and SEM (the common-factor restriction $\boldsymbol{\theta} = -\rho\boldsymbol{\beta}$) as testable restrictions (J. P. LeSage 2008).

Example 5.1 (IEEE 30 LMP regression, notebook §2a.4–2a.6)**.** Regress the 200-hour *average* LMP on betweenness, base-case demand, and congestion exposure (fraction of hours the bus is on a binding line) with $\mathbf{W} = \mathbf{A}$. Maximum likelihood gives

$$\hat{\rho}_{\text{SAR}} = -0.1056\ (z = -0.57), \qquad \hat{\lambda}_{\text{SEM}} = -0.4877,$$

with $\text{AIC}_{\text{SAR}} = 140.76$ and $\text{AIC}_{\text{SEM}} = 134.15$. The SAR effects (in €/MWh per unit of regressor) are

regressor	direct	indirect	total
betweenness	+4.331	−0.429	+3.902
demand	+0.052	−0.005	+0.047
exposure	−4.539	+0.450	−4.090

Neither $\hat{\rho}$ nor $\hat{\lambda}$ is decisively non-zero, and SEM has the lower AIC: averaged over 200 hours on a 30-bus system, the LMP spatial structure is faint, and what remains looks like nuisance correlation rather than a genuine price-spillover process. This is honest pedagogy — a single cross-section on $n = 30$ is statistically weak. The indirect effects are *negative* only because $\hat{\rho} < 0$; §8 shows the same machinery on a realistic benchmark where $\hat{\rho}$ becomes large and unambiguous.

6 The LM-diagnostic battery

How do we choose between SAR and SEM *before* estimating them? After fitting OLS $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, the Anselin, Bera, et al. (1996) battery asks two questions: (i) is there any spatial dependence left in the residuals, and (ii) is it of lag type ($\rho\mathbf{Wy}$) or error type ($\lambda\mathbf{Wu}$)?

- **Moran's I on residuals** is an omnibus test — it flags that *something* is wrong but not which model fits.
- **LM-lag** tests $H_0 : \rho = 0$ within SAR; **LM-error** tests $H_0 : \lambda = 0$ within SEM. Both are asymptotically χ^2_1.
- The basic forms are contaminated: LM-lag is biased upward under true error dependence and vice versa. The *robust* forms $\mathrm{LM}^\star_\rho, \mathrm{LM}^\star_\lambda$ net out the other direction.

Example 6.1 (IEEE 30 OLS diagnostics, notebook §2a.4)**.** On the 200-hour average LMP with $\mathbf{W} = \mathbf{A}$, the OLS fit gives $R^2 = 0.3016$ and a significant exposure coefficient (-4.404, $p = 0.016$), but its residuals fail both normality (Jarque–Bera $= 84.25$, $p < 10^{-4}$) and homoskedasticity (Breusch–Pagan $= 152.2$). Heavy-tailed, heteroskedastic residuals are precisely why we use *permutation* inference for Moran's I and ML/robust methods for the spatial models rather than trusting the OLS t-statistics. The accompanying `spat_diag` block reports the LM battery, which here does not decisively favour either lag or error dependence — consistent with the faint, non-robust $\hat{\rho}$ of Example 5.1.

Key idea

Decision rule: if only robust LM-lag is significant, fit SAR; if only robust LM-error is significant, fit SEM; if both are significant, fit SDM. Choosing the specification on these methodological grounds — not by R^2-shopping — is the point of the diagnostic battery. `spreg.OLS(..., spat_diag=True)` returns all five statistics in one call.

7 DG siting as a spatial regression

From describing prices to prescribing investment. Until now $\mathbf{y}$ was a price and spatial regression *described* the system. We now let $\mathbf{y}$ be a *suitability index* s_i for placing distributed generation (DG) at bus i, so spatial regression *prescribes* where to invest while accounting for network spillovers (Faria et al. 2016; Pereira and Saraiva 2011). Following the LaPSEE/UNESP siting tradition, suitability is a weighted blend of the price the DG would clear at, the demand it would relieve, the congestion it would avoid, and a renewable-resource proxy; the lab uses weights $(w_{\mathrm{LMP}}, w_{\mathrm{dem}}, w_{\mathrm{cong}}, w_{\mathrm{ren}}) = (0.35, 0.20, 0.20, 0.25)$. The four regressors are betweenness b_i (lines through which many shortest paths pass tend to relieve binding constraints), local demand density D_i, congestion exposure E_i (from Lab 2a), and renewable potential r_i.

Example 7.1 (IEEE 118 siting SAR, notebook §2b.2–2b.4)**.** On the larger IEEE 118 system with $\mathbf{W} = \mathbf{A}$, maximum likelihood gives a large, stable spatial-lag coefficient and a dominant congestion-exposure effect:

$$\hat{\rho} = +0.7269, \qquad \hat{\beta}_{\text{cong. exposure}} = +1.728\ (z = +12.09).$$

The LeSage–Pace effects (score units per unit of regressor) are

regressor	direct	indirect	total	% indirect
betweenness	+0.479	+0.901	+1.380	65.3%
demand density	+0.000	+0.000	+0.000	—
congestion exposure	+2.196	+4.130	**+6.327**	65.3%
renewable potential	+0.409	+0.768	+1.177	65.3%

A unit rise in congestion exposure at one bus raises that bus's own suitability by about 2.2 units (direct), while spillovers add about 4.1 units across the rest of the system — so ignoring spillovers *understates* the social value by a factor of roughly three. Ranking the IEEE 118 buses by average total effect versus by direct effect alone, 6 of the top-10 coincide and 4 are promoted by the spillover-aware ranking (buses $\{93, 95, 97, 98\}$, which sit downstream of high-exposure corridors), displacing $\{11, 16, 36, 84\}$ (Figure 3).

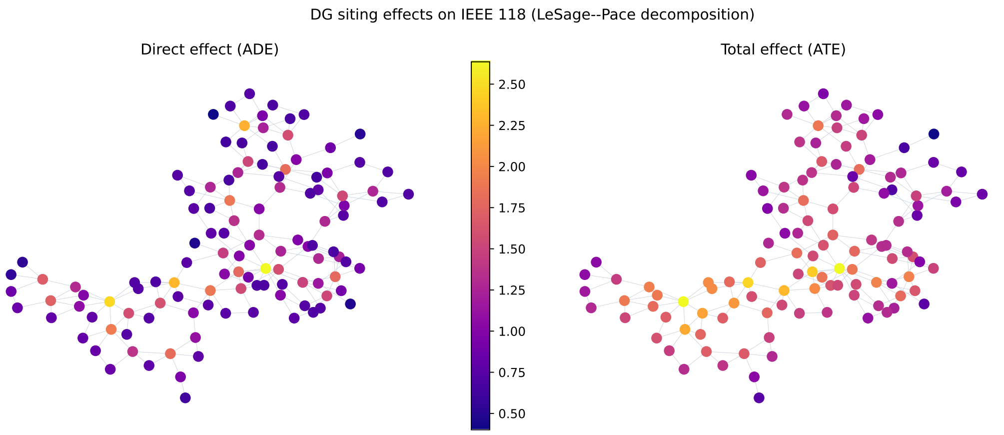

Figure 3: DG-siting effects on IEEE 118 (LeSage–Pace decomposition). Left: direct effect (ADE) per bus. Right: total effect (ATE) per bus, including spillovers. Buses that are network-pivotal gain most from the spillover-aware ranking. Notebook §2b.5.

Key idea

Rank candidate DG sites by the *average total effect*, not by the raw coefficient or the direct effect. The myopic ranking systematically undervalues network-pivotal buses by approximately the spillover share ($\sim$ 65% here). This is the policy-relevant output of the session and the bridge to Session 3, where agents will internalise exactly these spillovers through learning.

8 A reality check: RTS-GMLC

The IEEE benchmarks are pedagogically convenient but synthetic. As a sanity check we apply Lab 2a's machinery to **RTS-GMLC** (Barrows et al. 2020), NREL's 73-bus update of the classic RTS-96, with 105 lines, 155 generators (including solar PV and wind), hourly load for 8 760 hours, and bus coordinates. We solve a 168-hour DC-OPF panel on region 1.

Two findings matter. First, the LMP distribution is heavy-tailed and includes *negative prices*: across $73 \times 168 = 12\,264$ entries the mean is 19.51, median 24.94, range $[-3.38, +68.51]$ €/MWh, with 280 entries (2.3%) negative — a documented real-ISO phenomenon in renewable-rich systems (Edmunds et al. 2018) (Figure 4). Second, the spatial dependence that was faint on IEEE 30 is now dominant:

$$\hat{\rho}_{\text{IEEE 30}} = -0.106 \text{ (NS)}, \qquad \hat{\rho}_{\text{RTS-GMLC}} = \mathbf{+0.967} \text{ (strongly significant)},$$

and the mean Moran's I rises from +0.066 to +0.910.

Key idea

Benchmark size matters. On a small synthetic grid the spatial structure of LMPs is faint and statistically weak; on a realistic 73-bus benchmark it dominates. A single $\hat{\rho}$ from a single

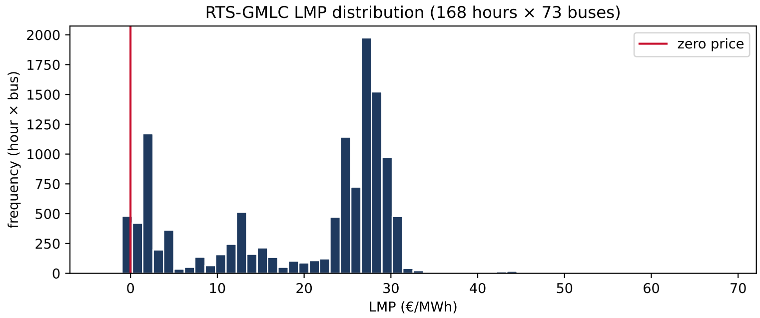

Figure 4: RTS-GMLC LMP distribution over 168 hours $\times$ 73 buses. The red line marks the zero-price level; 2.3% of entries are negative, reflecting renewable-curtailment hours. Notebook §3.2.

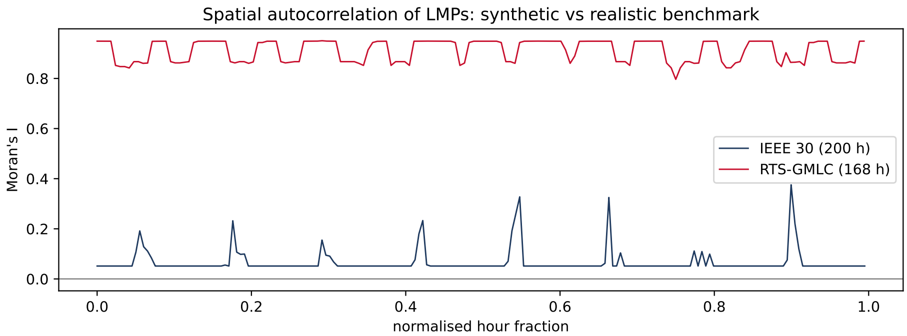

Figure 5: Per-hour Moran's I of nodal LMPs, IEEE 30 (navy) versus RTS-GMLC (red), with the hour axis normalised to each panel's length. The realistic benchmark sustains strong positive autocorrelation; the synthetic 30-bus system hovers near zero. Notebook §3.3.

$\mathbf{W}$ on a single benchmark is not a result — a defensible analysis reports a robustness table across systems and $\mathbf{W}$ choices (the notebook capstone builds the 3×4 version).

9 Limitations and the bridge to Session 3

Three honest limitations. (i) We used cross-sections; LMP data are inherently spatio-temporal, and panel SAR with bus and hour fixed effects, dynamic SAR, or spatial GARCH would be needed for inference on the full panel (J. LeSage and Pace 2009, Ch. 7). (ii) Identification is fragile under endogeneity and unobserved network constraints. (iii) The choice of $\mathbf{W}$ is a modelling decision, and LeSage's argument for SDM as a defensive default is precisely a hedge against getting it wrong.

The bridge to Session 3 is *economic*. The centralized DC-OPF that produced our LMPs assumes truthful bids and complete information. Both fail in real markets (California 2000–2001, Enron strategies). When they fail, the LMP signal becomes noisy and centralized clearing under-performs decentralized learning. In Session 3, agents observe their local PTDF rows, act on the very LMP we derived as a dual variable here, and communicate over the same $\mathbf{W}$ that defined spatial dependence in Session 1. The same $\mathbf{B}$, the same $\mathbf{H}$, the same $\mathbf{W}$ — three sessions, one mathematical object.

10 Exercises

The notebook `s02_spatial-prices-siting.ipynb` contains `TODO` + `self-check` cells for the starred exercises. Difficulty labels follow the LaPSEE Skill-B convention (apply / analyze / evaluate / create).

Exercise 10.1 (Verify the LMP decomposition, *routine · apply*)**.** Using the snapshot of Example 2.1, confirm that $\mathrm{LMP}_i - \lambda_{\text{energy}}$ equals the reported congestion component for buses 1–4, and that it is exactly zero at the slack bus. State the relative error.

Hint. Read `snap["lmp"]` and `snap["lambda_energy"]` from the notebook §1 output; the slack-bus congestion must be 0 to machine precision.

Exercise 10.2 (Binding lines and PTDF columns, *routine · apply*)**.** For each binding line in the snapshot (lines 0 and 20), report the bus at which $|H_{\ell,\cdot}|$ is largest and its value. Explain in one sentence why that bus carries a high congestion premium.

Hint. The notebook already computes this in cell §1.3; expected answers are bus 1 (0.833) on line 0 and bus 20 (0.575) on line 20.

Exercise 10.3 (Row standardisation of the four $\mathbf{W}$, *routine · apply*)**.** Row-standardise the four weight matrices of Example 3.1 and confirm $\sum_j \tilde{\mathbf{W}}_{ij} = 1$ for every non-island bus. Report the spectral radius $\mu_{\max}(\tilde{\mathbf{W}})$ for each and verify it equals 1.

Hint. Use `lapsee_s02.row_standardize(W)`; the four constructors are re-exported from Session 1.

Exercise 10.4 (Moran's I and the congestion regime, *routine · apply*)**.** Reproduce the correlation between $|I_t|$ and the per-hour binding-line count on the IEEE 30 panel ($\approx +0.867$). Then recompute I for the single *uncongested* hour with the fewest binding lines and confirm it is close to the null value $-1/(n-1)$.

Hint. Replicate cell §2a.2 and select the hour from `panel.attrs["binding_lines"]`.

Exercise 10.5 (Robustness of $\hat{\rho}$ across $\mathbf{W}$, *intermediate · analyze*)**. (Notebook Exercise 2.1.)** Re-fit the IEEE 30 SAR of Example 5.1 using the electrical-distance $\mathbf{W}_{\text{elec}} = 1/Z_{\text{eff}}$ instead of the binary adjacency, and report the new $\hat{\rho}$. Does the headline conclusion survive the $\mathbf{W}$ swap? Write 2–3 sentences interpreting what happens to $\hat{\rho}$ on the near-dense electrical-distance matrix.

Hint. Build $\mathbf{W}_{\text{elec}}$ with `lapsee_s02.W_electrical_distance(net)`, row-standardise, and re-run `spreg.ML_Lag`. The self-check asserts both $\hat{\rho}$ lie in $(-1, 1)$; note that the dense $\mathbf{W}_{\text{elec}}$ drives the estimate to the stationarity boundary.

Exercise 10.6 (ATE vs ADE ranking shift, *intermediate · analyze*)**. (Notebook Exercise 2.2.)** On IEEE 118, compute the symmetric difference of the top-10 buses ranked by average total effect and by average direct effect. Report the count and name 2–3 buses that the ATE ranking promotes but the ADE ranking misses, and explain the network reason.

Hint. Expected result: $\{93, 95, 97, 98\}$ appear only in the ATE top-10 and $\{11, 16, 36, 84\}$ only in the ADE top-10. The self-check asserts the two rankings are not identical.

Exercise 10.7 (Reading the synthetic-vs-realistic contrast, *synthesis · evaluate*)**. (Notebook Exercise 2.3.)** Fit a SAR on the mean RTS-GMLC LMP using betweenness and bus demand only, report $\hat{\rho}$, and write a short paragraph comparing the spillover structure with the IEEE 30 case. Argue *why* the spillovers are stronger on the larger benchmark, referencing the LMP decomposition of §2.3 and the role of the binding-constraint set. Conclude with one sentence on what you would report to a referee who saw only the IEEE 30 result.

Hint. Expected $\hat{\rho}_{\text{RTS}} \approx +0.967$ vs $\hat{\rho}_{\text{IEEE 30}} = -0.106$. Anchor the argument on the size of $\mathcal{B}$ (the binding set) and the smoothness of $\lambda\boldsymbol{\delta}$ across a larger network; cite Edmunds et al. (2018) and J. P. LeSage (2008).

11 Recommended pre- and post-reading

If you have time before class:

- Schweppe et al. (1988), the foundational treatment of spot pricing — read the introduction and the chapter deriving nodal prices.

- Litvinov (2010) for a clean, modern, ISO-oriented exposition of the LMP decomposition.
- J. LeSage and Pace (2009), Ch. 2–3, for SAR/SEM/SDM and the direct/indirect/total effects you will report in both labs.
- Anselin, Bera, et al. (1996) for the LM-diagnostic battery used to select a specification in Lab 2a.

After class, to go deeper:

- Hogan (1992) for the financial-transmission-rights and congestion-rent interpretation of LMP differences.
- Faria et al. (2016) and Pereira and Saraiva (2011) for the generation-siting framing behind Lab 2b.
- Edmunds et al. (2018) for empirical LMP variability (and negative prices) at the distribution level, the motivation for the RTS-GMLC mini-case.

References

Anselin, Luc (1995). "Local Indicators of Spatial Association — LISA". In: *Geographical Analysis* 27.2, pp. 93–115.

Anselin, Luc, Anil K. Bera, Raymond Florax, and Mann J. Yoon (1996). "Simple Diagnostic Tests for Spatial Dependence". In: *Regional Science and Urban Economics* 26.1, pp. 77–104.

Anselin, Luc and Sergio J. Rey (2014). *Modern Spatial Econometrics in Practice: A Guide to GeoDa, GeoDaSpace, and PySAL*. GeoDa Press.

Barrows, Clayton, Aaron Bloom, Ali Ehlen, Jussi Ikäheimo, Jennie Jorgenson, Dheepak Krishnamurthy, Jessica Lau, Brendan McBennett, Matthew O'Connell, Eugene Preston, Andrea Staid, Gord Stephen, and Jean-Paul Watson (2020). "The IEEE Reliability Test System: A Proposed 2019 Update". In: *IEEE Transactions on Power Systems*. Vol. 35. 1, pp. 119–127.

Dörfler, Florian, John W. Simpson-Porco, and Francesco Bullo (2018). "Electrical Networks and Algebraic Graph Theory: Models, Properties, and Applications". In: *Proceedings of the IEEE* 106.5, pp. 977–1005.

Edmunds, Calum, Waqquas Ahmed Bukhsh, Simon Gill, and Stuart Galloway (2018). *Locational Marginal Price Variability at Distribution Level: A Regional Study*. arXiv: `1804.04550`.

Faria, L. T., J. D. Melo, and A. Padilha-Feltrin (2016). "Spatial Analysis of Non-Technical Losses in Distribution Systems". In: *LaPSEE / UNESP Technical Report*.

Hogan, William W. (1992). "Contract Networks for Electric Power Transmission". In: *Journal of Regulatory Economics* 4.3, pp. 211–242.

Kelejian, Harry H. and Ingmar R. Prucha (1998). "A Generalized Spatial Two-Stage Least Squares Procedure for Estimating a Spatial Autoregressive Model with Autoregressive Disturbances". In: *The Journal of Real Estate Finance and Economics* 17.1, pp. 99–121.

Klein, Douglas J. and Milan Randić (1993). "Resistance Distance". In: *Journal of Mathematical Chemistry* 12.1, pp. 81–95.

Lee, Lung-Fei (2004). "Asymptotic Distributions of Quasi-Maximum Likelihood Estimators for Spatial Autoregressive Models". In: *Econometrica* 72.6, pp. 1899–1925.

LeSage, James and R. Kelley Pace (2009). *Introduction to Spatial Econometrics*. CRC Press.

LeSage, James P. (2008). "An Introduction to Spatial Econometrics". In: *Revue d'économie industrielle* 123, pp. 19–44.

Litvinov, Eugene (2010). "Design and Operation of the Locational Marginal Prices-Based Electricity Markets". In: *IET Generation, Transmission & Distribution* 4.2, pp. 315–323.

Moran, Patrick A. P. (1950). "Notes on Continuous Stochastic Phenomena". In: *Biometrika* 37.1/2, pp. 17–23.

Pereira, André J. C. and João T. Saraiva (2011). "Generation Expansion Planning (GEP) — A Long-Term Approach Using System Dynamics and Genetic Algorithms". In: *Energy* 36.8, pp. 5180–5199.

Schweppe, Fred C., Michael C. Caramanis, Richard D. Tabors, and Roger E. Bohn (1988). *Spot Pricing of Electricity*. Kluwer Academic Publishers.

Wang, Zhifang, Anna Scaglione, and Robert J. Thomas (2011). “Electrical Betweenness Approach for Vulnerability Assessment of Power Grids”. In: *Physica A: Statistical Mechanics and its Applications* 390.23, pp. 4692–4701.